

E9083776_1

The equation $x^4 + xy + y^3 + 7 = 0$ has no integer solutions

Size $h = 35$, length $l = 11.81$

Theorem 1. *The equation*

$$x^4 + xy + y^3 + 7 = 0 \tag{1}$$

has no integer solutions.

The proof uses eight polynomial identities of the form $H_k T_k = U_k^2 + d_k V_k^2$ on the curve (1), where $d_1 = -H_2$, $d_2 = -H_1$, $d_3 = -H_4$, $d_4 = -H_3$, $d_5 = -H_6$, $d_6 = -H_5$, $d_7 = -H_8$, and $d_8 = -H_7$, and the product formula for four Hilbert symbols. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Legendre symbol and of the Hilbert symbol.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. *Let p be an odd prime, and let a, b, r, c, t be integers with $a \neq 0$, $b \neq 0$ and $at = r^2 - bc^2$. If $p \nmid b$, and p does not divide both r and c , then $(a, b)_p = 1$.*

Proof. If $p \nmid a$, this is Theorem 2(ii) with $\alpha = \beta = 0$. If $p \mid a$, then $r^2 \equiv bc^2 \pmod{p}$. If $p \mid c$, then $p \mid r$, which is excluded. Hence $p \nmid c$, so $b \equiv (rc')^2 \pmod{p}$, where $cc' \equiv 1 \pmod{p}$, and $(b/p) = 1$ as $p \nmid b$. Theorem 2(ii) with $\beta = 0$ gives $(a, b)_p = (b/p)^{v_p(a)} = 1$. $\square$

Lemma 4. Let a, b, r, c, t be integers with $at = r^2 - bc^2 \neq 0$ and $b \neq 0$. Then $(a, b)_v = (t, b)_v$ for every v .

Proof. The triple $(r, s, z) = (1, c, r)$ is a non-zero rational solution of $z^2 = (at)r^2 + bs^2$, so $(at, b)_v = 1$. By Theorem 2(v), $(a, b)_v(t, b)_v = (at, b)_v = 1$, and both symbols are ± 1 . $\square$

Lemma 5. Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

Lemma 6. Let a, b, c be real numbers with $a > 0$, $c > 0$ and $256ac^3 > 27b^4$. Then $at^4 + bt + c > 0$ for every real number t .

Proof. For $t = 0$ this is clear. For $t \neq 0$, the inequality between the arithmetic and geometric means of at^4 , $c/3$, $c/3$ and $c/3$ gives $at^4 + c \geq 4(at^4c^3/27)^{1/4} = (256ac^3/27)^{1/4}|t| > |b||t| \geq -bt$. $\square$

2 The auxiliary polynomials

Put $F(x, y) = x^4 + xy + y^3 + 7$, so that (1) is the equation $F(x, y) = 0$. For the first identity, we define

$$\begin{aligned} H_1 &= -798x - 266y - 931, \\ T_1 &= 2154600x^4 - 718200x^3y - 60233040x^3 + 239400x^2y^2 \\ &\quad + 20995380x^2y - 168403536x^2 - 7277760xy^2 + 29213982xy - 116592588x \\ &\quad - 1247274y^2 + 4365459y - 41625864, \\ U_1 &= 7980x^3 - 229026x^2 - 421344x - 158802, \\ V_1 &= 8589, \\ G_1 &= -63680400x^2 + 1935884160x + 331774884, \end{aligned}$$

for the second identity, we define

$$\begin{aligned} H_2 &= 76x - 95y - 152, \\ T_2 &= 608000x^4 + 760000x^3y + 7341296x^3 + 950000x^2y^2 + 9509120x^2y \\ &\quad + 12477984x^2 + 13786400xy^2 + 17382720xy - 41616688x + 49301200y^2 \\ &\quad - 78881920y + 56211072, \\ U_2 &= 9500x^3 + 71364x^2 + 2964x - 31084, \\ V_2 &= 5000, \\ G_2 &= -90250000x^2 - 1309708000x - 4683614000, \end{aligned}$$

for the third identity, we define

$$\begin{aligned} H_3 &= 80x^2 - 8x - 10y - 16, \\ T_3 &= 8356659200000x^8 - 3681894400000x^7 + 1044582400000x^6y \\ &\quad + 4003449920000x^6 - 355778560000x^5y - 935632832000x^5 \\ &\quad + 130572800000x^4y^2 + 673769864000x^4y - 601964310400x^4 \\ &\quad - 31415040000x^3y^2 - 137054429600x^3y + 280330837608x^3 \end{aligned}$$

$$\begin{aligned}
& + 107194289000x^2y^2 + 49729871040x^2y - 429361595792x^2 \\
& - 12695382800xy^2 - 795830240xy + 61432390136x + 26385553400y^2 \\
& - 42216885440y - 111982837296, \\
U_3 &= 25856000x^5 - 6988800x^4 + 3258180x^3 - 52996x^2 - 2224084x + 219356, \\
V_3 &= 358100, \\
G_3 &= -1305728000000x^4 + 314150400000x^3 - 1071942890000x^2 + 126953828000x \\
& - 263855534000,
\end{aligned}$$

for the fourth identity, we define

$$\begin{aligned}
H_4 &= -14y - 28, \\
T_4 &= 56x^3 + 56x^2y^2 - 112x^2y + 168x^2 - 56xy^2 + 112xy - 434x - 210y^2 \\
& + 420y - 875, \\
U_4 &= 28x^3 - 14x^2 - 56x - 56, \\
V_4 &= 7, \\
G_4 &= -784x^2 + 784x + 2940,
\end{aligned}$$

for the fifth identity, we define

$$\begin{aligned}
H_5 &= 28x^2 + 56x - 84y + 21, \\
T_5 &= 280000x^8 + 2195200x^7 + 840000x^6y + 14206192x^6 + 4905600x^5y \\
& + 20515712x^5 + 2520000x^4y^2 + 32177376x^4y - 24790780x^4 \\
& + 9676800x^3y^2 - 14046816x^3y - 1147552x^3 + 75288528x^2y^2 \\
& - 99442140x^2y - 234331755x^2 - 199975104xy^2 - 19888848xy + 47711412x \\
& + 45157392y^2 + 11289348y - 1077668172, \\
U_5 &= 2800x^5 + 13776x^4 + 97944x^3 - 83860x^2 - 36645x - 62622, \\
V_5 &= 301266, \\
G_5 &= -211680000x^4 - 812851200x^3 - 6324236352x^2 + 16797908736x - 3793220928,
\end{aligned}$$

for the sixth identity, we define

$$\begin{aligned}
H_6 &= 2x - y, \\
T_6 &= 16x^2 + 8xy + 4x + 4y^2 - 84, \\
U_6 &= 2x^2 + 8x - 7, \\
V_6 &= 1, \\
G_6 &= -4,
\end{aligned}$$

for the seventh identity, we define

$$\begin{aligned}
H_7 &= 28x^2 - 21xy - 98x - 77y - 175, \\
T_7 &= 2721600x^5 + 2041200x^4y + 50008140x^4 + 1530900x^3y^2 \\
& + 53282880x^3y + 276693144x^3 + 12916260x^2y^2 + 119241108x^2y \\
& - 567490364x^2 + 15600816xy^2 + 247697436xy + 1105390384x + 193767840y^2 \\
& - 591622472y + 1980738452,
\end{aligned}$$

$$\begin{aligned}
U_7 &= 41034x^3 + 68068x^2 + 87304x - 5670y^3 - 93870y^2 - 279188y + 313460, \\
V_7 &= 458136, \\
G_7 &= 76204800x^3 - 550278036x^2 + 32148900xy - 3215874312x - 32148900y^3 \\
&\quad - 1064485800y^2 - 11977568820y - 63555200100,
\end{aligned}$$

for the eighth identity, we define

$$\begin{aligned}
H_8 &= 2x - y, \\
T_8 &= 16x^4 + 8x^3y + 148x^3 + 4x^2y^2 + 72x^2y + 136x^2 + 36xy^2 + 50xy \\
&\quad + 4x + 25y^2 - 77, \\
U_8 &= 2x^3 + 17x^2 + 8x, \\
V_8 &= 1, \\
G_8 &= -4x^2 - 36x - 25,
\end{aligned}$$

for the auxiliary polynomials H_1 and H_2 together, we define

$$\begin{aligned}
\lambda_1 &= 40x^3 + 268x^2 - 456xy - 6x + 152y^2 - 532y - 1013, \\
\mu_1 &= -112x^3 + 3080x^2 + 9380x + 8050, \\
\nu_1 &= 40432,
\end{aligned}$$

for the auxiliary polynomials H_3 and H_4 together, we define

$$\begin{aligned}
\lambda_2 &= 180006400x^3y + 357200200x^3 - 125422080x^2y - 249446960x^2 \\
&\quad + 22500800xy^2 - 25259136xy - 140241122x - 13427680y^2 + 21484288y \\
&\quad + 102262671, \\
\mu_2 &= 1028608000x^5 - 819558400x^4 - 533532760x^3 + 474789072x^2 \\
&\quad + 107171878x - 97598237, \\
\nu_2 &= 225008000x - 134276800,
\end{aligned}$$

for the auxiliary polynomials H_5 and H_6 together, we define

$$\begin{aligned}
\lambda_3 &= 24064x^4 - 12032x^3y + 208376x^3 + 22148x^2y + 162204x^2 - 36096xy^2 \\
&\quad + 83400xy + 272970x + 138636y^2 + 34659y - 125325, \\
\mu_3 &= -336896x^5 - 2075024x^4 - 14180768x^3 - 10551240x^2 - 7591752x \\
&\quad + 11255139, \\
\nu_3 &= -3032064x + 11645424,
\end{aligned}$$

for the auxiliary polynomials H_7 and H_8 together, we define

$$\begin{aligned}
\lambda_4 &= 23408x^3 + 170370x^2 - 76844x + 62467, \\
\mu_4 &= -491568x^4 - 4069338x^3 + 655424x^2y + 11472146x^2 + 327712xy^2 \\
&\quad + 11324600xy + 10267481x + 5662300y^2 - 4809959, \\
\nu_4 &= 327712x + 5662300,
\end{aligned}$$

and, for the signs, we also use

$$Q_1 = 636804x^2 - 212268xy + 1556632x + 70756y^2 - 247646y + 866761,$$

$$\begin{aligned}
M_1 &= 18821096x^4 - 508169592x^3 - 1835056860x^2 - 2140899670x - 675206819, \\
E_1 &= 135173111472x^2 + 335429572912x + 183985623948, \\
Q_2 &= 5776x^2 + 7220xy - 14079x + 9025y^2 - 14440y + 23104, \\
M_2 &= 857375x^4 + 438976x^3 - 1947956x^2 + 3895912x + 2489817, \\
E_2 &= 156385200x^2 - 299738300x + 625540800, \\
Q_3 &= 6400x^4 - 1280x^3 + 800x^2y - 2496x^2 - 80xy + 356x + 100y^2 - 160y + 256, \\
M_3 &= 512000x^6 - 153600x^5 - 290840x^4 + 68928x^3 + 57568x^2 - 7744x + 2904, \\
E_3 &= 1920000x^4 - 384000x^3 - 748800x^2 + 116800x + 76800, \\
L_4 &= y, \\
Q_4 &= y^3 + 7, \\
\Delta_4 &= 256y^9 + 5376y^6 - 27y^4 + 37632y^3 + 87808, \\
Q_5 &= 784x^4 + 3136x^3 + 2352x^2y + 4312x^2 + 4704xy + 9408x + 7056y^2 + 1764y \\
&\quad + 441, \\
M_5 &= 21952x^6 + 131712x^5 + 905520x^4 + 570752x^3 + 629748x^2 + 222264x \\
&\quad + 4158189, \\
E_5 &= 16595712x^4 + 66382848x^3 + 91276416x^2 + 248935680x + 9335088, \\
Q_6 &= 4x^2 + 2xy + x + y^2, \\
M_6 &= x^4 + 8x^3 + 2x^2 + 7, \\
E_6 &= 12x^2 + 4x, \\
Q_8 &= 4x^2 + 2xy + x + y^2, \\
M_8 &= x^4 + 8x^3 + 2x^2 + 7, \\
E_8 &= 12x^2 + 4x.
\end{aligned}$$

Expanding both sides, one checks the identities

$$H_1T_1 - U_1^2 + H_2V_1^2 = G_1F, \quad (2)$$

$$H_2T_2 - U_2^2 + H_1V_2^2 = G_2F, \quad (3)$$

$$H_3T_3 - U_3^2 + H_4V_3^2 = G_3F, \quad (4)$$

$$H_4T_4 - U_4^2 + H_3V_4^2 = G_4F, \quad (5)$$

$$H_5T_5 - U_5^2 + H_6V_5^2 = G_5F, \quad (6)$$

$$H_6T_6 - U_6^2 + H_5V_6^2 = G_6F, \quad (7)$$

$$H_7T_7 - U_7^2 + H_8V_7^2 = G_7F, \quad (8)$$

$$H_8T_8 - U_8^2 + H_7V_8^2 = G_8F, \quad (9)$$

$$\lambda_1H_1 + \mu_1H_2 + \nu_1F = 7 \cdot 19^2, \quad (10)$$

$$\lambda_2H_3 + \mu_2H_4 + \nu_2F = 2^2 \cdot 5^2 \cdot 7 \cdot 11^2 \cdot 43^2, \quad (11)$$

$$\lambda_3H_5 + \mu_3H_6 + \nu_3F = 3^4 \cdot 7 \cdot 373^2, \quad (12)$$

$$\lambda_4H_7 + \mu_4H_8 + \nu_4F = 3^8 \cdot 5^4 \cdot 7, \quad (13)$$

$$-18821096F = Q_1H_1 - M_1, \quad (14)$$

$$283024Q_1 = (-212268x + 141512y - 247646)^2 + E_1. \quad (15)$$

$$-857375F = Q_2H_2 - M_2, \quad (16)$$

$$36100Q_2 = (7220x + 18050y - 14440)^2 + E_2. \quad (17)$$

$$-1000F = Q_3H_3 - M_3, \quad (18)$$

$$400Q_3 = (800x^2 - 80x + 200y - 160)^2 + E_3. \quad (19)$$

$$F = x^4 + L_4x + Q_4, \quad (20)$$

$$\Delta_4 = 256Q_4^3 - 27L_4^4, \quad (21)$$

$$-592704F = Q_5H_5 - M_5, \quad (22)$$

$$28224Q_5 = (2352x^2 + 4704x + 14112y + 1764)^2 + E_5. \quad (23)$$

$$-F = Q_6H_6 - M_6, \quad (24)$$

$$4Q_6 = (2x + 2y)^2 + E_6. \quad (25)$$

$$-F = Q_8H_8 - M_8, \quad (26)$$

$$4Q_8 = (2x + 2y)^2 + E_8. \quad (27)$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) . Put

$$\begin{aligned} A_1 &= -d_2 = H_1, & B_1 &= -d_1 = H_2, & T_{A_1} &= T_1, & T_{B_1} &= T_2, \\ A_2 &= -d_4 = H_3, & B_2 &= -d_3 = H_4, & T_{A_2} &= T_3, & T_{B_2} &= T_4, \\ A_3 &= -d_6 = H_5, & B_3 &= -d_5 = H_6, & T_{A_3} &= T_5, & T_{B_3} &= T_6, \\ A_4 &= -d_8 = H_7, & B_4 &= -d_7 = H_8, & T_{A_4} &= T_7, & T_{B_4} &= T_8. \end{aligned}$$

By (2)–(9), after multiplying by squares of the constants c in $d_k = cH_{k'}$ if necessary, we obtain

$$A_1T_{A_1} = U_1^2 - B_1V_1^2, \quad B_1T_{B_1} = U_2^2 - A_1V_2^2. \quad (28)$$

$$A_2T_{A_2} = U_3^2 - B_2V_3^2, \quad B_2T_{B_2} = U_4^2 - A_2V_4^2. \quad (29)$$

$$A_3T_{A_3} = U_5^2 - B_3V_5^2, \quad B_3T_{B_3} = U_6^2 - A_3V_6^2. \quad (30)$$

$$A_4T_{A_4} = U_7^2 - B_4V_7^2, \quad B_4T_{B_4} = U_8^2 - A_4V_8^2. \quad (31)$$

Step 1: signs. *The sign of H_1 .* Since $F(x, y) = 0$, (14) gives $Q_1H_1 = M_1$, and M_1 and E_1 depend only on x . We show that $M_1 > 0$ and $E_1 > 0$. If $x \geq 31$, write $x = 31 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_1 and E_1 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_1 > 0$ and $E_1 > 0$. It remains to consider $-1 \leq x \leq 30$. For $x \in \{-1, 2, 5, 8, 11, 14, 17, 20, 23, 26, 29\}$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with such x . For $x \in \{3, 4, 9, 13, 18, 19, 24, 28\}$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with such x . For $x \in \{1, 12, 15, 22\}$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with such x . For $x \in \{6, 7, 10, 21\}$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , as one checks for the 11 residues of y modulo 11, so (1) has no solution with such x . For $x = 0$ and $x = 27$, we have $F(x, y) \not\equiv 0 \pmod{13}$ for every y , as one checks for the 13 residues of y modulo 13, so (1) has no solution with such x . For $x \in \{16, 25, 30\}$, we have $F(x, y) \not\equiv 0 \pmod{19}$ for every y , as one checks for the 19 residues of y modulo 19, so (1) has no solution with such x . Hence $M_1 > 0$ and $E_1 > 0$, so (15) gives $283024Q_1 \geq E_1 > 0$, and $Q_1H_1 = M_1 > 0$ gives $H_1 > 0$.

The sign of H_2 . Since $F(x, y) = 0$, (16) gives $Q_2H_2 = M_2$, and M_2 and E_2 depend only on x . We show that $M_2 > 0$ and $E_2 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -3$, write $x = -3 - z$ with $z \geq 0$. In both cases, expanding M_2 and E_2 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_2 > 0$ and $E_2 > 0$. It remains to consider $-2 \leq x \leq 0$. For $x = 0$ we have $M_2 > 0$ and $E_2 > 0$. For $x = -1$, we have $F(x, y) \not\equiv 0$

(mod 3) for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with this x . For $x = -2$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . Hence $M_2 > 0$ and $E_2 > 0$, so (17) gives $36100Q_2 \geq E_2 > 0$, and $Q_2H_2 = M_2 > 0$ gives $H_2 > 0$.

The sign of H_3 . Since $F(x, y) = 0$, (18) gives $Q_3H_3 = M_3$, and M_3 and E_3 depend only on x . We show that $M_3 > 0$ and $E_3 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_3 and E_3 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_3 > 0$ and $E_3 > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_3 > 0$ and $E_3 > 0$. Hence $M_3 > 0$ and $E_3 > 0$, so (19) gives $400Q_3 \geq E_3 > 0$, and $Q_3H_3 = M_3 > 0$ gives $H_3 > 0$.

The sign of H_4 . Suppose that $H_4 \leq 0$. Since $H_4 = -14y - 28$, this means $y \geq -2$. By (20), F is $x^4 + L_4x + Q_4$, where L_4 and Q_4 depend only on y . If $y \geq 1$, write $y = 1 + z$ with $z \geq 0$; expanding Q_4 and Δ_4 as polynomials in z gives non-negative coefficients and positive constant terms, so $Q_4 > 0$ and $\Delta_4 > 0$. For $y = -1$ and $y = 0$ we also have $Q_4 > 0$ and $\Delta_4 > 0$. For $y = -2$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every x , so (1) has no solution with this y . Hence $Q_4 > 0$ and $\Delta_4 > 0$, so Lemma 6 with $a = 1$, $b = L_4$, $c = Q_4$ and $t = x$ gives $F(x, y) > 0$, a contradiction. Hence $H_4 > 0$.

The sign of H_5 . Since $F(x, y) = 0$, (22) gives $Q_5H_5 = M_5$, and M_5 and E_5 depend only on x . We show that $M_5 > 0$ and $E_5 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -4$, write $x = -4 - z$ with $z \geq 0$. In both cases, expanding M_5 and E_5 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_5 > 0$ and $E_5 > 0$. It remains to consider $-3 \leq x \leq 0$. For $x = 0$ we have $M_5 > 0$ and $E_5 > 0$. For $x = -1$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with this x . For $x = -2$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . For $x = -3$, we have $F(x, y) \not\equiv 0 \pmod{17}$ for every y , as one checks for the 17 residues of y modulo 17, so (1) has no solution with this x . Hence $M_5 > 0$ and $E_5 > 0$, so (23) gives $28224Q_5 \geq E_5 > 0$, and $Q_5H_5 = M_5 > 0$ gives $H_5 > 0$.

The sign of H_6 . Since $F(x, y) = 0$, (24) gives $Q_6H_6 = M_6$, and M_6 and E_6 depend only on x . We show that $M_6 > 0$ and $E_6 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -8$, write $x = -8 - z$ with $z \geq 0$. In both cases, expanding M_6 and E_6 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_6 > 0$ and $E_6 > 0$. It remains to consider $-7 \leq x \leq 0$. For $x = -1$ we have $M_6 > 0$ and $E_6 > 0$. For $x = -7$ and $x = -4$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with such x . For $x = -6$ and $x = -2$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with such x . For $x = -5$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , as one checks for the 11 residues of y modulo 11, so (1) has no solution with this x . For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{13}$ for every y , as one checks for the 13 residues of y modulo 13, so (1) has no solution with this x . For $x = -3$, we have $F(x, y) \not\equiv 0 \pmod{17}$ for every y , as one checks for the 17 residues of y modulo 17, so (1) has no solution with this x . Hence $M_6 > 0$ and $E_6 > 0$, so (25) gives $4Q_6 \geq E_6 > 0$, and $Q_6H_6 = M_6 > 0$ gives $H_6 > 0$.

The sign of H_8 . Since $F(x, y) = 0$, (26) gives $Q_8H_8 = M_8$, and M_8 and E_8 depend only on x . We show that $M_8 > 0$ and $E_8 > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -8$, write $x = -8 - z$ with $z \geq 0$. In both cases, expanding M_8 and E_8 as polynomials in z gives non-negative coefficients and positive constant terms, so $M_8 > 0$ and $E_8 > 0$. It remains to consider $-7 \leq x \leq 0$. For $x = -1$ we have $M_8 > 0$ and $E_8 > 0$. For $x = -7$ and $x = -4$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with such x . For $x = -6$ and $x = -2$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with such x . For $x = -5$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , as one checks for the 11 residues of y modulo 11, so (1) has no solution with this x . For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{13}$ for every y , as

one checks for the 13 residues of y modulo 13, so (1) has no solution with this x . For $x = -3$, we have $F(x, y) \not\equiv 0 \pmod{17}$ for every y , as one checks for the 17 residues of y modulo 17, so (1) has no solution with this x . Hence $M_8 > 0$ and $E_8 > 0$, so (27) gives $4Q_8 \geq E_8 > 0$, and $Q_8 H_8 = M_8 > 0$ gives $H_8 > 0$.

By these signs, $A_1 > 0$, $B_1 > 0$, $A_2 > 0$, $B_2 > 0$, $A_3 > 0$, $B_3 > 0$, and $B_4 > 0$.

Hence Theorem 2(i) gives $(A_1, B_1)_\infty = 1$ (as $A_1 > 0$), $(A_2, B_2)_\infty = 1$ (as $A_2 > 0$), $(A_3, B_3)_\infty = 1$ (as $A_3 > 0$), and $(A_4, B_4)_\infty = 1$ (as $B_4 > 0$).

Step 2: primes outside S . Let $S = \{2, 3, 5, 7, 11, 19, 43, 101, 373, 409, 797, 3581\}$ be the set of primes dividing 2, the constant 1 in the $d_k = cH_{k'}$, $V_1 = 8589$, $V_2 = 5000$, $V_3 = 358100$, $V_4 = 7$, $V_5 = 301266$, $V_6 = 1$, $V_7 = 458136$, and $V_8 = 1$, and the right-hand sides of (10), (11), (12), and (13). Let $p \notin S$ be an odd prime and $j \in \{1, 2, 3, 4\}$; we show that $(A_j, B_j)_p = 1$. If $p \nmid A_j B_j$, this follows from Theorem 2(ii) with $\alpha = \beta = 0$. For $j = 1$: If $p \mid A_1$, then $p \mid H_1$, so $p \nmid H_2$ by (10) and $p \nmid B_1$; as $p \nmid V_1$, Lemma 3 with $(a, b, r, c, t) = (A_1, B_1, U_1, V_1, T_{A_1})$ and (28) gives $(A_1, B_1)_p = 1$; if $p \mid B_1$, then $p \mid H_2$, so $p \nmid H_1$ by (10) and $p \nmid A_1$; as $p \nmid V_2$, Lemma 3 with $(a, b, r, c, t) = (B_1, A_1, U_2, V_2, T_{B_1})$ and (28) gives $(B_1, A_1)_p = 1$, which equals $(A_1, B_1)_p$ by Theorem 2(v). For $j = 2$: If $p \mid A_2$, then $p \mid H_3$, so $p \nmid H_4$ by (11) and $p \nmid B_2$; as $p \nmid V_3$, Lemma 3 with $(a, b, r, c, t) = (A_2, B_2, U_3, V_3, T_{A_2})$ and (29) gives $(A_2, B_2)_p = 1$; if $p \mid B_2$, then $p \mid H_4$, so $p \nmid H_3$ by (11) and $p \nmid A_2$; as $p \nmid V_4$, Lemma 3 with $(a, b, r, c, t) = (B_2, A_2, U_4, V_4, T_{B_2})$ and (29) gives $(B_2, A_2)_p = 1$, which equals $(A_2, B_2)_p$ by Theorem 2(v). For $j = 3$: If $p \mid A_3$, then $p \mid H_5$, so $p \nmid H_6$ by (12) and $p \nmid B_3$; as $p \nmid V_5$, Lemma 3 with $(a, b, r, c, t) = (A_3, B_3, U_5, V_5, T_{A_3})$ and (30) gives $(A_3, B_3)_p = 1$; if $p \mid B_3$, then $p \mid H_6$, so $p \nmid H_5$ by (12) and $p \nmid A_3$; as $p \nmid V_6$, Lemma 3 with $(a, b, r, c, t) = (B_3, A_3, U_6, V_6, T_{B_3})$ and (30) gives $(B_3, A_3)_p = 1$, which equals $(A_3, B_3)_p$ by Theorem 2(v). For $j = 4$: If $p \mid A_4$, then $p \mid H_7$, so $p \nmid H_8$ by (13) and $p \nmid B_4$; as $p \nmid V_7$, Lemma 3 with $(a, b, r, c, t) = (A_4, B_4, U_7, V_7, T_{A_4})$ and (31) gives $(A_4, B_4)_p = 1$; if $p \mid B_4$, then $p \mid H_8$, so $p \nmid H_7$ by (13) and $p \nmid A_4$; as $p \nmid V_8$, Lemma 3 with $(a, b, r, c, t) = (B_4, A_4, U_8, V_8, T_{B_4})$ and (31) gives $(B_4, A_4)_p = 1$, which equals $(A_4, B_4)_p$ by Theorem 2(v).

Step 3: the primes in S . By Step 1, the A_j and B_j are non-zero. If $T_{A_j} \neq 0$, then (28)–(31) and Lemma 4 give $(A_j, B_j)_v = (T_{A_j}, B_j)_v$, and if $T_{B_j} \neq 0$, they give $(A_j, B_j)_v = (B_j, A_j)_v = (T_{B_j}, A_j)_v = (A_j, T_{B_j})_v$, using Theorem 2(v). For a non-zero integer X and a prime p , we write $X \in p^{2i}(\pm)$ or $X \in p^{2i+1}(\pm)$, when p is odd, if $X = p^j u$ with $p \nmid u$, $(u/p) = \pm 1$, and j even or odd, respectively; and we write $X \in 2^{2i} \cdot u$ or $X \in 2^{2i+1} \cdot u$ if $X = 2^j u'$ with $u' \equiv u \pmod{8}$ and j even or odd. If $X \not\equiv 0 \pmod{p^e}$ and $v_p(X) \leq e - 1$, or $v_p(X) \leq e - 3$ if $p = 2$, then this class is determined by X modulo p^e . By Theorem 2(ii) and (iii), the classes of X and Y determine $(X, Y)_p$.

The prime 2. Checking the 64^2 residue pairs modulo 64, we find that (1) has exactly 96 solutions modulo 64, and that at each of them, for each j , one of the cases listed for $p = 2$ and j in Table 1 holds; moreover, the combination of these cases is one of the combinations listed there. Each case gives the stated value of $(A_j, B_j)_2$. Hence $\sigma_2 \in V_2$, with V_2 as in Step 4.

The prime 3. Lifting the solutions modulo 3 step by step to solutions modulo $19683 = 3^9$, we find that (1) has exactly 13122 solutions modulo 19683, and that at each of them, for each j , one of the cases listed for $p = 3$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_3$. Hence $\sigma_3 \in V_3$, with V_3 as in Step 4.

The prime 5. Checking the 5^2 residue pairs modulo 5, we find that (1) has exactly 4 solutions modulo 5, and that at each of them, for each j , one of the cases listed for $p = 5$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_5$. Hence $\sigma_5 \in V_5$, with V_5 as in Step 4.

The prime 7. Checking the 49^2 residue pairs modulo 49, we find that (1) has exactly 35 solutions modulo 49, and that at each of them, for each j , one of the cases listed for $p = 7$ and j in Table 1 holds; moreover, the combination of these cases is one of the combinations listed there. Each case gives the stated value of $(A_j, B_j)_7$. Hence $\sigma_7 \in V_7$, with V_7 as in Step 4.

The prime 11. Checking the 11^2 residue pairs modulo 11, we find that (1) has exactly 11 solutions modulo 11, and that at each of them, for each j , one of the cases listed for $p = 11$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_{11}$. Hence $\sigma_{11} \in V_{11}$, with V_{11} as in Step 4.

The prime 19. Checking the 19^2 residue pairs modulo 19, we find that (1) has exactly 18 solutions modulo 19, and that at each of them, for each j , one of the cases listed for $p = 19$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_{19}$. Hence $\sigma_{19} \in V_{19}$, with V_{19} as in Step 4.

The prime 43. Checking the 43^2 residue pairs modulo 43, we find that (1) has exactly 56 solutions modulo 43, and that at each of them, for each j , one of the cases listed for $p = 43$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_{43}$. Hence $\sigma_{43} \in V_{43}$, with V_{43} as in Step 4.

The prime 101. Checking the 101^2 residue pairs modulo 101, we find that (1) has exactly 112 solutions modulo 101, and that at each of them, for each j , one of the cases listed for $p = 101$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_{101}$. Hence $\sigma_{101} \in V_{101}$, with V_{101} as in Step 4.

The prime 373. Checking the 373^2 residue pairs modulo 373, we find that (1) has exactly 392 solutions modulo 373, and that at each of them, for each j , one of the cases listed for $p = 373$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_{373}$. Hence $\sigma_{373} \in V_{373}$, with V_{373} as in Step 4.

The prime 409. Checking the 409^2 residue pairs modulo 409, we find that (1) has exactly 379 solutions modulo 409, and that at each of them, for each j , one of the cases listed for $p = 409$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_{409}$. Hence $\sigma_{409} \in V_{409}$, with V_{409} as in Step 4.

The prime 797. Checking the 797^2 residue pairs modulo 797, we find that (1) has exactly 822 solutions modulo 797, and that at each of them, for each j , one of the cases listed for $p = 797$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_{797}$. Hence $\sigma_{797} \in V_{797}$, with V_{797} as in Step 4.

The prime 3581. Checking the 3581^2 residue pairs modulo 3581, we find that (1) has exactly 3611 solutions modulo 3581, and that at each of them, for each j , one of the cases listed for $p = 3581$ and j in Table 1 holds. Each case gives the stated value of $(A_j, B_j)_{3581}$. Hence $\sigma_{3581} \in V_{3581}$, with V_{3581} as in Step 4.

Step 4: contradiction. For $p \in S$ and $p = \infty$, put

$$\sigma_p = ((A_1, B_1)_p, (A_2, B_2)_p, (A_3, B_3)_p, (A_4, B_4)_p).$$

By Theorem 2(iv), applied to each j , and by Step 2, the componentwise product of the σ_p with $p \in S$ is σ_∞ , which is $(1, 1, 1, 1)$ by Step 1. By Step 3, $\sigma_p \in V_p$, where

$$\begin{aligned} V_2 &= \{(1, 1, 1, 1), (1, 1, 1, -1), \\ &\quad (1, -1, 1, -1), (-1, -1, 1, 1), \\ &\quad (-1, -1, 1, -1)\}, \\ V_3 &= \{(1, 1, 1, 1)\}, \\ V_5 &= \{(1, 1, 1, 1)\}, \\ V_7 &= \{(1, -1, 1, 1), (1, -1, -1, -1), \\ &\quad (-1, 1, 1, 1), (-1, 1, -1, -1)\}, \\ V_{11} &= \{(1, 1, 1, 1)\}, \\ V_{19} &= \{(1, 1, 1, 1)\}, \\ V_{43} &= \{(1, 1, 1, 1)\}, \end{aligned}$$

$$\begin{aligned}
V_{101} &= \{(1, 1, 1, 1)\}, \\
V_{373} &= \{(1, 1, 1, 1)\}, \\
V_{409} &= \{(1, 1, 1, 1)\}, \\
V_{797} &= \{(1, 1, 1, 1)\}, \\
V_{3581} &= \{(1, 1, 1, 1)\}.
\end{aligned}$$

The componentwise products of one vector from each of V_3 , V_5 , V_7 , V_{11} , V_{19} , V_{43} , V_{101} , V_{373} , V_{409} , V_{797} , and V_{3581} are $(1, -1, 1, 1)$, $(1, -1, -1, -1)$, $(-1, 1, 1, 1)$, and $(-1, 1, -1, -1)$, so σ_2 would have to be one of $(1, -1, 1, 1)$, $(1, -1, -1, -1)$, $(-1, 1, 1, 1)$, and $(-1, 1, -1, -1)$, but none of them is in V_2 . This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.
- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.

Table 1: The cases in Step 3. In each case, the symbol $(A_j, B_j)_p$ is computed as the stated symbol, whose entries lie in the stated classes; an entry “any” means that the other entry is a unit square, so that Lemma 5 applies.

	symbol	first entry	second entry	value
$p = 2$, solutions modulo 64				
$j = 1$				
1	(A_1, B_1)	$2^{2i} \cdot 1$	$2^{2i} \cdot 1$	1
2	(A_1, B_1)	$2^{2i} \cdot 1$	$2^{2i} \cdot 5$	1
3	(A_1, B_1)	$2^{2i} \cdot 3$	$2^{2i} \cdot 1$	1
4	(A_1, B_1)	$2^{2i} \cdot 3$	$2^{2i} \cdot 3$	-1
5	(A_1, B_1)	$2^{2i} \cdot 3$	$2^{2i} \cdot 5$	1
6	(A_1, B_1)	$2^{2i} \cdot 3$	$2^{2i} \cdot 7$	-1
7	(A_1, B_1)	$2^{2i} \cdot 5$	$2^{2i} \cdot 3$	1
8	(A_1, B_1)	$2^{2i} \cdot 5$	$2^{2i} \cdot 7$	1
9	(A_1, B_1)	$2^{2i} \cdot 7$	$2^{2i} \cdot 3$	-1
$j = 2$				
1	(A_2, B_2)	$2^{2i+1} \cdot 1$	$2^{2i} \cdot 5$	-1
2	(A_2, B_2)	$2^{2i+1} \cdot 1$	$2^{2i+1} \cdot 1$	1
3	(A_2, B_2)	$2^{2i+1} \cdot 1$	$2^{2i+1} \cdot 5$	-1
4	(A_2, B_2)	$2^{2i+1} \cdot 3$	$2^{2i+1} \cdot 3$	-1
5	(A_2, B_2)	$2^{2i+1} \cdot 3$	$2^{2i+1} \cdot 7$	1
6	(A_2, B_2)	$2^{2i+1} \cdot 5$	$2^{2i+1} \cdot 1$	-1
7	(A_2, B_2)	$2^{2i+1} \cdot 5$	$2^{2i+1} \cdot 5$	1
8	(A_2, B_2)	$2^{2i+1} \cdot 7$	$2^{2i} \cdot 5$	-1
9	(A_2, B_2)	$2^{2i+1} \cdot 7$	$2^{2i+1} \cdot 3$	1
10	(A_2, B_2)	$2^{2i+1} \cdot 7$	$2^{2i+1} \cdot 7$	-1
$j = 3$				
1	(A_3, B_3)	$2^{2i} \cdot 1$	$2^{2i} \cdot 1$	1
2	(A_3, B_3)	$2^{2i} \cdot 1$	$2^{2i} \cdot 3$	1
3	(A_3, B_3)	$2^{2i} \cdot 1$	$2^{2i} \cdot 5$	1
4	(A_3, B_3)	$2^{2i} \cdot 1$	$2^{2i} \cdot 7$	1
5	(A_3, B_3)	$2^{2i} \cdot 1$	$2^{2i+1} \cdot 1$	1
6	(A_3, B_3)	$2^{2i} \cdot 1$	$2^{2i+1} \cdot 3$	1
7	(A_3, B_3)	$2^{2i} \cdot 1$	$2^{2i+1} \cdot 5$	1

	symbol	first entry	second entry	value
8	(A_3, B_3)	$2^{2i} \cdot 1$	$2^{2i+1} \cdot 7$	1
9	(A_3, B_3)	$2^{2i} \cdot 5$	$2^{2i} \cdot 1$	1
10	(A_3, B_3)	$2^{2i} \cdot 5$	$2^{2i} \cdot 3$	1
11	(A_3, B_3)	$2^{2i} \cdot 5$	$2^{2i} \cdot 5$	1
12	(A_3, B_3)	$2^{2i} \cdot 5$	$2^{2i} \cdot 7$	1
$j = 4$				
1	(A_4, B_4)	$2^{2i} \cdot 3$	$2^{2i} \cdot 3$	-1
2	(A_4, B_4)	$2^{2i} \cdot 3$	$2^{2i+1} \cdot 1$	-1
3	(A_4, B_4)	$2^{2i} \cdot 3$	$2^{2i+1} \cdot 5$	-1
4	(A_4, B_4)	$2^{2i} \cdot 7$	$2^{2i} \cdot 1$	1
5	(A_4, B_4)	$2^{2i} \cdot 7$	$2^{2i} \cdot 3$	-1
6	(A_4, B_4)	$2^{2i} \cdot 7$	$2^{2i} \cdot 5$	1
7	(A_4, B_4)	$2^{2i} \cdot 7$	$2^{2i} \cdot 7$	-1
8	(A_4, B_4)	$2^{2i} \cdot 7$	$2^{2i+1} \cdot 3$	-1
9	(A_4, B_4)	$2^{2i} \cdot 7$	$2^{2i+1} \cdot 7$	-1
10	(T_{A_4}, B_4)	$2^{2i} \cdot 1$	$2^{2i} \cdot 5$	1
11	(T_{A_4}, B_4)	$2^{2i} \cdot 5$	$2^{2i} \cdot 1$	1
combinations of cases ($j = 1, j = 2, j = 3, j = 4$) and the vector σ_2				
$(1, 5, 9, 4) \mapsto (1, 1, 1, 1) \quad (2, 9, 11, 6) \mapsto (1, 1, 1, 1)$				
$(3, 4, 4, 7) \mapsto (1, -1, 1, -1) \quad (3, 8, 6, 8) \mapsto (1, -1, 1, -1)$				
$(3, 8, 8, 9) \mapsto (1, -1, 1, -1) \quad (4, 3, 1, 11) \mapsto (-1, -1, 1, 1)$				
$(5, 10, 2, 1) \mapsto (1, -1, 1, -1) \quad (6, 6, 3, 10) \mapsto (-1, -1, 1, 1)$				
$(7, 2, 12, 7) \mapsto (1, 1, 1, -1) \quad (8, 7, 10, 5) \mapsto (1, 1, 1, -1)$				
$(9, 1, 5, 2) \mapsto (-1, -1, 1, -1) \quad (9, 1, 7, 3) \mapsto (-1, -1, 1, -1)$				
$p = 3$, solutions modulo 19683				
$j = 1$				
1	$3 \nmid A_1 B_1$			1
2	(A_1, B_1)	$3^{2i}(+)$	$3^{2i}(-)$	1
3	(A_1, B_1)	$3^{2i}(+)$	$3^{2i}(+)$	1
4	(A_1, B_1)	$3^{2i}(+)$	$3^{2i+1}(-)$	1
5	(A_1, B_1)	$3^{2i}(+)$	$3^{2i+1}(+)$	1
6	(A_1, T_{B_1})	$3^{2i}(+)$	$3^{2i}(-)$	1
$j = 2$				
1	(A_2, B_2)	$3^{2i}(-)$	$3^{2i}(+)$	1
2	(A_2, B_2)	$3^{2i}(+)$	$3^{2i}(+)$	1
3	(A_2, B_2)	$3^{2i+1}(-)$	$3^{2i}(+)$	1
4	(A_2, B_2)	$3^{2i+1}(+)$	$3^{2i}(+)$	1
5	(T_{A_2}, B_2)	$3^{2i}(-)$	$3^{2i}(+)$	1
6	(T_{A_2}, B_2)	$3^{2i}(+)$	$3^{2i}(+)$	1
$j = 3$				
1	(A_3, B_3)	$3^{2i}(-)$	$3^{2i}(-)$	1
2	(A_3, B_3)	$3^{2i}(-)$	$3^{2i}(+)$	1
3	(A_3, B_3)	$3^{2i}(+)$	$3^{2i}(-)$	1
4	(A_3, B_3)	$3^{2i}(+)$	$3^{2i}(+)$	1
5	(A_3, B_3)	$3^{2i}(+)$	$3^{2i+1}(-)$	1
6	(A_3, B_3)	$3^{2i}(+)$	$3^{2i+1}(+)$	1
7	(A_3, B_3)	$3^{2i+1}(-)$	$3^{2i}(+)$	1
8	(A_3, B_3)	$3^{2i+1}(-)$	$3^{2i+1}(+)$	1
9	(A_3, B_3)	$3^{2i+1}(+)$	$3^{2i}(+)$	1
10	(A_3, B_3)	$3^{2i+1}(+)$	$3^{2i+1}(-)$	1
11	(A_3, T_{B_3})	$3^{2i}(+)$	$3^{2i}(+)$	1
12	(T_{A_3}, B_3)	$3^{2i}(+)$	$3^{2i}(+)$	1
13	(T_{A_3}, B_3)	$3^{2i+1}(-)$	$3^{2i}(+)$	1
$j = 4$				
1	$3 \nmid A_4 B_4$			1
2	(A_4, B_4)	$3^{2i}(-)$	$3^{2i}(-)$	1
3	(A_4, B_4)	$3^{2i}(-)$	$3^{2i}(+)$	1

	symbol	first entry	second entry	value
4	(A_4, B_4)	$3^{2i}(+)$	$3^{2i}(-)$	1
5	(A_4, B_4)	$3^{2i}(+)$	$3^{2i}(+)$	1
6	(A_4, B_4)	$3^{2i}(+)$	$3^{2i+1}(-)$	1
7	(A_4, B_4)	$3^{2i}(+)$	$3^{2i+1}(+)$	1
8	(A_4, B_4)	$3^{2i+1}(-)$	$3^{2i}(+)$	1
9	(A_4, B_4)	$3^{2i+1}(-)$	$3^{2i+1}(+)$	1
10	(A_4, B_4)	$3^{2i+1}(+)$	$3^{2i}(+)$	1
11	(A_4, B_4)	$3^{2i+1}(+)$	$3^{2i+1}(-)$	1
12	(A_4, T_{B_4})	$3^{2i}(+)$	$3^{2i}(+)$	1
13	(T_{A_4}, B_4)	$3^{2i}(+)$	$3^{2i}(+)$	1
14	(T_{A_4}, B_4)	$3^{2i+1}(+)$	$3^{2i}(+)$	1

$p = 5$, solutions modulo 5

$j = 1$				
1	$5 \nmid A_1 B_1$			1
2	(A_1, B_1)	$5^{2i}(+)$	any	1
3	(T_{A_1}, B_1)	$5^{2i}(+)$	$5^{2i}(+)$	1
$j = 2$				
1	$5 \nmid A_2 B_2$			1
$j = 3$				
1	$5 \nmid A_3 B_3$			1
2	(A_3, T_{B_3})	$5^{2i}(+)$	$5^{2i}(+)$	1
$j = 4$				
1	$5 \nmid A_4 B_4$			1
2	(T_{A_4}, B_4)	$5^{2i}(+)$	any	1

$p = 7$, solutions modulo 49

$j = 1$				
1	(A_1, B_1)	$7^{2i+1}(-)$	$7^{2i}(+)$	1
2	(A_1, B_1)	$7^{2i+1}(+)$	$7^{2i}(-)$	-1
3	(A_1, B_1)	$7^{2i+1}(+)$	$7^{2i}(+)$	1
$j = 2$				
1	(A_2, B_2)	$7^{2i}(-)$	$7^{2i+1}(-)$	-1
2	(A_2, B_2)	$7^{2i}(-)$	$7^{2i+1}(+)$	-1
3	(A_2, B_2)	$7^{2i}(+)$	$7^{2i+1}(-)$	1
4	(A_2, B_2)	$7^{2i}(+)$	$7^{2i+1}(+)$	1
$j = 3$				
1	(A_3, B_3)	$7^{2i+1}(-)$	$7^{2i}(-)$	-1
2	(A_3, B_3)	$7^{2i+1}(-)$	$7^{2i}(+)$	1
3	(A_3, B_3)	$7^{2i+1}(-)$	$7^{2i+1}(-)$	-1
4	(A_3, B_3)	$7^{2i+1}(+)$	$7^{2i}(+)$	1
$j = 4$				
1	(A_4, B_4)	$7^{2i+1}(-)$	$7^{2i}(-)$	-1
2	(A_4, B_4)	$7^{2i+1}(-)$	$7^{2i}(+)$	1
3	(A_4, B_4)	$7^{2i+1}(-)$	$7^{2i+1}(-)$	-1
combinations of cases ($j = 1, j = 2, j = 3, j = 4$) and the vector σ_7				
$(1, 1, 1, 1) \mapsto (1, -1, -1, -1) \quad (2, 3, 3, 3) \mapsto (-1, 1, -1, -1)$				
$(2, 3, 4, 2) \mapsto (-1, 1, 1, 1) \quad (2, 4, 4, 2) \mapsto (-1, 1, 1, 1)$				
$(3, 2, 2, 2) \mapsto (1, -1, 1, 1)$				

$p = 11$, solutions modulo 11

$j = 1$				
1	$11 \nmid A_1 B_1$			1
2	(A_1, T_{B_1})	$11^{2i}(+)$	$11^{2i}(-)$	1
3	(A_1, T_{B_1})	$11^{2i}(+)$	$11^{2i}(+)$	1
4	(T_{A_1}, B_1)	$11^{2i}(+)$	$11^{2i}(+)$	1
$j = 2$				
1	$11 \nmid A_2 B_2$			1
2	(T_{A_2}, B_2)	$11^{2i}(+)$	any	1
3	(T_{A_2}, B_2)	$11^{2i}(+)$	$11^{2i}(+)$	1

	symbol	first entry	second entry	value
$j = 3$				
1	$11 \nmid A_3 B_3$			1
2	(A_3, B_3)	any	$11^{2i}(+)$	1
3	(A_3, T_{B_3})	$11^{2i}(+)$	$11^{2i}(-)$	1
4	(A_3, T_{B_3})	$11^{2i}(+)$	$11^{2i}(+)$	1
$j = 4$				
1	$11 \nmid A_4 B_4$			1
2	(A_4, T_{B_4})	$11^{2i}(+)$	$11^{2i}(-)$	1
3	(A_4, T_{B_4})	$11^{2i}(+)$	$11^{2i}(+)$	1
$p = 19$, solutions modulo 19				
$j = 1$				
1	(T_{A_1}, B_1)	$19^{2i}(+)$	any	1
$j = 2$				
1	$19 \nmid A_2 B_2$			1
2	(A_2, T_{B_2})	$19^{2i}(+)$	$19^{2i}(+)$	1
$j = 3$				
1	$19 \nmid A_3 B_3$			1
2	(A_3, T_{B_3})	$19^{2i}(+)$	$19^{2i}(-)$	1
$j = 4$				
1	$19 \nmid A_4 B_4$			1
2	(A_4, T_{B_4})	$19^{2i}(+)$	$19^{2i}(-)$	1
3	(T_{A_4}, B_4)	$19^{2i}(+)$	$19^{2i}(+)$	1
$p = 43$, solutions modulo 43				
$j = 1$				
1	$43 \nmid A_1 B_1$			1
2	(A_1, T_{B_1})	$43^{2i}(+)$	$43^{2i}(-)$	1
3	(A_1, T_{B_1})	$43^{2i}(+)$	$43^{2i}(+)$	1
$j = 2$				
1	$43 \nmid A_2 B_2$			1
2	(A_2, T_{B_2})	$43^{2i}(+)$	$43^{2i}(-)$	1
3	(A_2, T_{B_2})	$43^{2i}(+)$	$43^{2i}(+)$	1
4	(T_{A_2}, B_2)	$43^{2i}(-)$	$43^{2i}(+)$	1
5	(T_{A_2}, B_2)	$43^{2i}(+)$	any	1
$j = 3$				
1	$43 \nmid A_3 B_3$			1
2	(A_3, T_{B_3})	$43^{2i}(+)$	$43^{2i}(+)$	1
3	(T_{A_3}, B_3)	$43^{2i}(+)$	$43^{2i}(+)$	1
$j = 4$				
1	$43 \nmid A_4 B_4$			1
2	(A_4, T_{B_4})	$43^{2i}(+)$	$43^{2i}(+)$	1
3	(T_{A_4}, B_4)	$43^{2i}(-)$	$43^{2i}(+)$	1
$p = 101$, solutions modulo 101				
$j = 1$				
1	$101 \nmid A_1 B_1$			1
2	(T_{A_1}, B_1)	$101^{2i}(+)$	$101^{2i}(+)$	1
$j = 2$				
1	$101 \nmid A_2 B_2$			1
2	(A_2, T_{B_2})	$101^{2i}(+)$	$101^{2i}(-)$	1
3	(T_{A_2}, B_2)	$101^{2i}(-)$	$101^{2i}(+)$	1
$j = 3$				
1	$101 \nmid A_3 B_3$			1
2	(A_3, T_{B_3})	$101^{2i}(+)$	$101^{2i}(-)$	1
3	(A_3, T_{B_3})	$101^{2i}(+)$	$101^{2i}(+)$	1
4	(T_{A_3}, B_3)	$101^{2i}(-)$	$101^{2i}(+)$	1
$j = 4$				
1	$101 \nmid A_4 B_4$			1
2	(A_4, B_4)	any	$101^{2i}(+)$	1

	symbol	first entry	second entry	value
3	(A_4, T_{B_4})	$101^{2i}(+)$	$101^{2i}(-)$	1
4	(T_{A_4}, B_4)	$101^{2i}(+)$	$101^{2i}(+)$	1
$p = 373$, solutions modulo 373				
$j = 1$				
1	$373 \nmid A_1 B_1$			1
2	(A_1, T_{B_1})	$373^{2i}(+)$	$373^{2i}(-)$	1
3	(T_{A_1}, B_1)	$373^{2i}(-)$	$373^{2i}(+)$	1
4	(T_{A_1}, B_1)	$373^{2i}(+)$	$373^{2i}(+)$	1
$j = 2$				
1	$373 \nmid A_2 B_2$			1
2	(A_2, T_{B_2})	$373^{2i}(+)$	$373^{2i}(-)$	1
3	(A_2, T_{B_2})	$373^{2i}(+)$	$373^{2i}(+)$	1
4	(T_{A_2}, B_2)	$373^{2i}(-)$	$373^{2i}(+)$	1
$j = 3$				
1	$373 \nmid A_3 B_3$			1
2	(T_{A_3}, B_3)	$373^{2i}(+)$	any	1
3	(T_{A_3}, B_3)	$373^{2i}(+)$	$373^{2i}(+)$	1
$j = 4$				
1	$373 \nmid A_4 B_4$			1
2	(A_4, T_{B_4})	$373^{2i}(+)$	$373^{2i}(+)$	1
3	(T_{A_4}, B_4)	$373^{2i}(+)$	$373^{2i}(+)$	1
$p = 409$, solutions modulo 409				
$j = 1$				
1	$409 \nmid A_1 B_1$			1
2	(T_{A_1}, B_1)	$409^{2i}(+)$	$409^{2i}(+)$	1
$j = 2$				
1	$409 \nmid A_2 B_2$			1
2	(T_{A_2}, B_2)	$409^{2i}(-)$	$409^{2i}(+)$	1
$j = 3$				
1	$409 \nmid A_3 B_3$			1
2	(T_{A_3}, B_3)	$409^{2i}(-)$	$409^{2i}(+)$	1
$j = 4$				
1	$409 \nmid A_4 B_4$			1
$p = 797$, solutions modulo 797				
$j = 1$				
1	$797 \nmid A_1 B_1$			1
2	(A_1, T_{B_1})	$797^{2i}(+)$	$797^{2i}(+)$	1
3	(T_{A_1}, B_1)	$797^{2i}(-)$	$797^{2i}(+)$	1
$j = 2$				
1	$797 \nmid A_2 B_2$			1
2	(T_{A_2}, B_2)	$797^{2i}(+)$	$797^{2i}(+)$	1
$j = 3$				
1	$797 \nmid A_3 B_3$			1
2	(A_3, B_3)	any	$797^{2i}(+)$	1
3	(A_3, T_{B_3})	$797^{2i}(+)$	$797^{2i}(-)$	1
4	(A_3, T_{B_3})	$797^{2i}(+)$	$797^{2i}(+)$	1
5	(T_{A_3}, B_3)	$797^{2i}(+)$	$797^{2i}(+)$	1
$j = 4$				
1	$797 \nmid A_4 B_4$			1
2	(A_4, T_{B_4})	$797^{2i}(+)$	$797^{2i}(-)$	1
3	(A_4, T_{B_4})	$797^{2i}(+)$	$797^{2i}(+)$	1
4	(T_{A_4}, B_4)	$797^{2i}(-)$	$797^{2i}(+)$	1
5	(T_{A_4}, B_4)	$797^{2i}(+)$	$797^{2i}(+)$	1
$p = 3581$, solutions modulo 3581				
$j = 1$				
1	$3581 \nmid A_1 B_1$			1

	symbol	first entry	second entry	value
2	(A_1, T_{B_1})	$3581^{2i}(+)$	$3581^{2i}(+)$	1
3	(T_{A_1}, B_1)	$3581^{2i}(+)$	$3581^{2i}(+)$	1
$j = 2$				
1	$3581 \nmid A_2 B_2$			1
2	(A_2, T_{B_2})	$3581^{2i}(+)$	$3581^{2i}(-)$	1
3	(A_2, T_{B_2})	$3581^{2i}(+)$	$3581^{2i}(+)$	1
4	(T_{A_2}, B_2)	$3581^{2i}(+)$	$3581^{2i}(-)$	1
$j = 3$				
1	$3581 \nmid A_3 B_3$			1
2	(A_3, T_{B_3})	$3581^{2i}(+)$	$3581^{2i}(-)$	1
3	(T_{A_3}, B_3)	$3581^{2i}(+)$	$3581^{2i}(+)$	1
$j = 4$				
1	$3581 \nmid A_4 B_4$			1
2	(A_4, T_{B_4})	$3581^{2i}(+)$	$3581^{2i}(-)$	1
3	(T_{A_4}, B_4)	$3581^{2i}(-)$	$3581^{2i}(+)$	1